

KCH-MedRank: Knowledge-Constrained Learning-to-Rank for Biomedical Evidence Retrieval

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Abstract

Biomedical retrievers routinely place gold evidence passages inside the candidate pool and then fail to pull them into the top 10. The passages are not irrelevant. Pairwise query-passage scoring simply has no way to see the higher-order medical relations that connect them—a passage about the thyroid hormone analog DITPA and a passage about levothyroxine share almost no surface lexical overlap, yet both reside in the same Medical Subject Headings (MeSH) subtree for thyroid hormones. We introduce KCH-MedRank, a knowledge-constrained learning-to-rank framework for biomedical evidence reranking that gives the ranker access to retrieval scores, MeSH hierarchy signals, biomedical entity overlap, PrimeKG knowledge-graph relations, and an optional local hypergraph. On BioASQ, KCH-MedRank beats a true MedCPT Cross-Encoder baseline on Recall@10 (+0.0157, $p = 0.0076$) and runs $18.6\times$ faster at reranking time (reranking-stage latency under cached features, excluding candidate generation). The central finding is that flat biomedical knowledge features explain most aggregate gains. Hypergraph structure provides targeted early-rank improvements under indirect-evidence conditions, but the full model does not significantly exceed the flat knowledge baseline at the standard top-10 evaluation depth ($\Delta\text{Recall@10} = +0.0020$, n.s.). Controlled ablation indicates that domain-grounded annotations are critical for productive graph-based reranking, although a full factorial graph-by-knowledge design would further isolate topology effects. We categorize 648 of 661 rescued passages by dominant activated knowledge mechanism—MeSH hierarchy paths account for 75.9% of them, shared entity clusters for 23.9%, and PrimeKG relations for the remainder. Supplementary diagnostics on PubMedQA and BEIR NFCorpus indicate that supervised reranking with retrieval features generalizes, while full knowledge-augmented generalization remains to be tested.

Keywords: Biomedical evidence retrieval; learning-to-rank; medical knowledge graph; MeSH hierarchy; hypergraph diffusion; interpretable reranking

1 Introduction

Ask a biomedical search engine “What is the function of DITPA?” and it retrieves dozens of passages about thyroid hormone analogs. A clinician scanning those results spots the connection immediately. DITPA is a thyroid hormone analog. Its mechanism sits under the Medical Subject Headings (MeSH) tree node for thyroid hormones. A standard retrieval pipeline scores each passage against the query by lexical overlap or embedding similarity, and it places the most relevant evidence at rank 54. The passage is sitting in the candidate pool. Pairwise scoring cannot see it.

Across 943 held-out BioASQ test questions, a strong Hybrid RRF retriever leaves every gold evidence passage outside the top 10 for over 6% of queries where that evidence does exist somewhere in the top 100. This happens because biomedical evidence lives in higher-order relations. Questions, passages, MeSH concepts, biomedical entities, documents, and knowledge-base entries

connect through pathways that pairwise query-passage comparisons cannot trace. A passage about DITPA and a passage about levothyroxine share no surface lexical overlap. They sit far apart in embedding space. They belong to the same MeSH subtree, and a pairwise scoring function never asks about that.

Existing biomedical retrievers [7] and medical RAG systems [20, 21] score each passage independently against the query. They do not ask whether two candidate passages share a MeSH ancestor. They do not check whether the biomedical entities in a question form a dense cluster with those of a low-ranked passage. GraphRAG-style approaches [5, 12, 19] do apply graph structure to retrieval, but their edges come from lexical overlap or embedding similarity rather than domain-grounded concept annotations such as MeSH, biomedical entity linking, or curated knowledge graphs like PrimeKG. No existing work isolates whether raw graph topology, stripped of medical knowledge, helps or actively hurts biomedical evidence ranking.

We introduce KCH-MedRank, a knowledge-constrained learning-to-rank framework for biomedical evidence reranking. The method starts from a reproducible candidate pool built with BM25, dense retrieval, and Reciprocal Rank Fusion (RRF). It enriches each query’s local candidate neighborhood with four categories of features: (1) retrieval scores from lexical, dense, and fused sources; (2) biomedical semantic features from MeSH hierarchy annotations and entity overlap metrics; (3) relation features from the PrimeKG biomedical knowledge graph; and (4) optional structural features from a local hypergraph that captures n-ary concept interactions that pairwise graphs cannot preserve. All features are fed into query-grouped LambdaMART—a gradient-boosted tree-based learning-to-rank algorithm [2]—yielding a transparent, feature-attributable reranker. The framework is designed as a focused evidence selection module that can serve as a retrieval stage in medical RAG pipelines, though the experiments in this paper center on retrieval and reranking metrics.

As controlled baselines we use a flat knowledge-feature model that removes the hypergraph entirely, and a pairwise graph decomposition that breaks multi-node hyperedges into anchor-based edges. The core experimental design isolates three questions. **RQ1:** Do flat biomedical knowledge features alone improve on retrieval signals? **RQ2:** Does hypergraph structure provide additional benefit, and under what conditions? **RQ3:** Are domain-grounded concept annotations—MeSH, entity, PrimeKG—necessary for graph-based biomedical reranking, or would generic graph topology built from lexical or embedding similarity suffice?

Our contributions are the following. First, a knowledge-constrained learning-to-rank framework that integrates retrieval, biomedical semantic, MeSH hierarchy, entity, relation, and optional hypergraph features for interpretable evidence reranking. Second, a controlled ablation showing that flat biomedical knowledge features explain most aggregate gains; hypergraph structure provides targeted but modest early-rank benefit concentrated in questions lacking direct MeSH overlap and in multi-evidence scenarios; and removing domain-grounded concept annotations from the hypergraph degrades ranking, suggesting that generic graph topology built without medical knowledge is insufficient for productive graph-based reranking. Third, on BioASQ, statistically significant Recall@10 improvement over a true MedCPT Cross-Encoder baseline (+0.0157, $p = 0.0076$) with 18.6× faster reranking (reranking-stage latency under cached features, excluding candidate generation), plus supplementary diagnostics on PubMedQA and BEIR NFCorpus. Fourth, a diagnostic analysis categorizing 648 of 661 rescued gold passages by dominant activated knowledge mechanism, with MeSH hierarchy paths accounting for 75.9% of rescues, shared entity clusters for 23.9%, and PrimeKG relations for the remainder.

We position this work as a focused evidence selection module for medical RAG rather than a complete clinical QA system. The method does not replace clinical judgment or guarantee answer correctness. The failure analysis reveals cases where gold passages remain beyond the reach of our knowledge features, and the hypergraph benefit over flat knowledge features is modest and

concentrated at early ranks ($k \leq 5$). These limitations are discussed transparently, along with directions for broader knowledge-graph coverage and generation-level evaluation.

2 Related Work

2.1 Medical Retrieval-Augmented Question Answering

Medical RAG systems [20] have shown that retrieval quality dominates downstream QA performance, often mattering more than the choice of generator model. Iterative medical RAG [21] tries to expand evidence coverage by generating follow-up retrieval queries. Benchmarking studies systematically evaluate corpora, retrievers, and backbone models [20]. But every system in this space shares a basic architecture: each passage gets scored independently against the query. Evidence selection stays pairwise. None of these pipelines has a mechanism for reranking evidence using the higher-order biomedical relations that connect passages to each other and to structured knowledge. A passage that shares a MeSH ancestor with the query but no surface lexical overlap remains invisible no matter how many retrieval rounds you run.

2.2 Biomedical Semantic Retrieval

MedCPT [7] trains a dual-encoder retriever and a cross-encoder reranker on large-scale PubMed search logs. These are strong neural baselines that capture biomedical semantics far better than generic retrievers. The BioASQ benchmark [9, 17] provides the standard evaluation framework. PubMedQA [6] and BEIR NFCorpus [16] offer complementary testbeds. All of these semantic models operate on isolated query-passage pairs. They learn from the statistical patterns of PubMed click logs. They do not use freely available structured biomedical knowledge: MeSH hierarchies, biomedical entity graphs, curated relation databases like PrimeKG. Our question is whether adding these knowledge sources as ranking features improves evidence reranking beyond what semantic models alone achieve.

2.3 Graph and Hypergraph Approaches to RAG

A growing number of systems apply graph structure to retrieved information. GraphRAG [5] organizes passages into entity-relationship graphs and performs local-to-global search for summarization. Medical Graph RAG [19] grounds medical responses by linking documents to credible sources and controlled vocabularies. HyperGraphRAG [12] uses hyperedges for passage clustering and argues that n-ary relations capture complex information structures more naturally than pairwise edges. Cross-granularity HGRAG [18] models entities and passages at different retrieval granularities. Medical HyperRAG [22] builds a ClinBridge hypergraph over clinical records for patient-centered QA.

These methods share a limitation: they apply graph structure without domain-grounded concept annotations. Edges are built from lexical overlap, embedding similarity, or co-occurrence statistics. These are proxies; they may or may not reflect genuine biomedical relationships. None of these works systematically compares graph-based reranking with and without medical knowledge annotations. Our work provides that comparison. We compare the same hypergraph architecture with and without MeSH, entity, and PrimeKG annotations, and show that domain knowledge constraints are essential for productive graph-based reranking, while also acknowledging that a full 2×2 factorial design—with/without graph $\times$ with/without medical knowledge—would provide stronger causal isolation.

2.4 Controlled Biomedical Knowledge

MeSH [13] is a hierarchically organized controlled vocabulary maintained by the National Library of Medicine, providing standardized annotations for all PubMed-indexed literature. PrimeKG [3] integrates multimodal precision-medicine relations across diseases, proteins, drugs, and phenotypes. Both are public and widely used in biomedical information retrieval. Their use as features in controlled LTR reranking settings has not been widely studied. Our framework uses MeSH hierarchy features as the primary structured knowledge signal and PrimeKG as an auxiliary source. The design choice we want to highlight is not which knowledge base to use but how to wire structured biomedical knowledge as features inside a learning-to-rank model, so that after training we can inspect which knowledge pathways drove each ranking decision.

3 Method

3.1 Problem Formulation

We address evidence reranking for biomedical question answering. Given a medical question q , a biomedical corpus C , and structured knowledge sources K that span the MeSH hierarchy, biomedical entity annotations, and PrimeKG relations, the task produces a ranked evidence list E where gold passages appear as early as possible. A first-stage retriever returns a candidate pool $D_q = \{d_1, \dots, d_M\} \subset C$. The reranker learns a scoring function $f(q, d)$ over D_q that minimizes a listwise ranking loss $\mathcal{L}$:

$$\min_f \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathcal{L}(f(q, \cdot), R_q),$$

where R_q encodes binary gold relevance judgments ($y_{q,d} \in \{0, 1\}$) for the candidate passages of q . The loss enforces $f(q, d_i) > f(q, d_j)$ for all pairs where d_i is relevant and d_j is not. Evaluation uses Recall@ k , MRR@ k , and nDCG@ k at the passage level. Answer generation is treated as a downstream diagnostic.

The challenge is that pairwise scoring functions miss the higher-order biomedical relations connecting questions and evidence. Our framework addresses this by constructing an interpretable feature representation $\phi(q, d)$ that encodes retrieval signals, biomedical semantic scores, MeSH hierarchy features, entity overlap features, relation features, and optional hypergraph structural features.

3.2 Overall Framework

KCH-MedRank builds a final evidence ranking in four connected stages: multi-source candidate retrieval, flat biomedical knowledge feature extraction, optional local hypergraph construction with diffusion-based structural features, and query-group LambdaMART learning-to-rank. The framework is designed so that each component can be ablated independently. Removing the hypergraph yields the flat knowledge baseline. Stripping MeSH and entity annotations from the hypergraph yields the no-knowledge variant. Breaking multi-node hyperedges into anchor-based pairwise edges produces the graph decomposition baseline.

3.2.1 First-Stage Candidate Generation

The reranker can only reorder passages already in the candidate pool, so the first-stage ceiling matters. We use four retrieval sources: BM25 as the lexical baseline [15], dense biomedical retrieval

Figure 1. Overview of KCH-MedRank

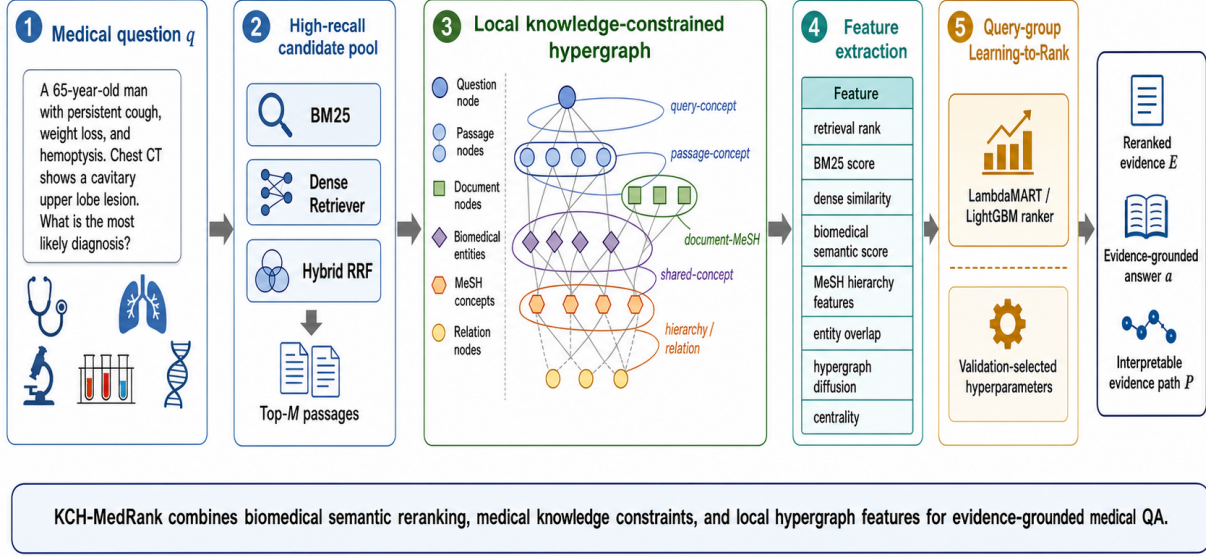


Figure 1: Overview of KCH-MedRank. The method first builds a high-recall candidate pool, constructs a local knowledge-constrained hypergraph around each question and candidate set, and uses hypergraph and biomedical semantic features in query-group learning-to-rank.

as the semantic baseline, fielded BM25 with title, text, and MeSH expansion fields, and MedCPT dual-encoder retrieval [7] as an additional biomedical semantic source. These are fused via weighted Reciprocal Rank Fusion [4]:

$$\text{RRF}(q, d) = \sum_{s \in S} \frac{w_s}{k + r_s(q, d)},$$

where $r_s(q, d)$ is the rank of passage d under source s , w_s is the source weight, and $k = 60$. The selected w122 configuration uses BM25 and dense retrieval at weight 1.0, fielded BM25 at 1.2, and MedCPT dual-encoder at 0.2. This setting is chosen solely to maximize validation Recall@100, never test performance, because reranking can only improve passages that the first stage places in the candidate pool.

3.2.2 Flat Biomedical Knowledge Features

The core of KCH-MedRank is the set of biomedical knowledge features computed for each query-passage pair without any graph construction. These features encode what a clinician would check when comparing a question to a candidate passage: shared concepts, hierarchical relationships, and known biomedical relations.

Entity features measure concept-level alignment. Let E_q and E_d be the sets of biomedical entities extracted from q and d :

$$\phi_{\text{ent-jacc}} = \frac{|E_q \cap E_d|}{|E_q \cup E_d|}, \quad \phi_{\text{ent-cov}} = \frac{|E_q \cap E_d|}{|E_q|}.$$

MeSH features operate at two granularities. Let M_q and M_d be the sets of MeSH descriptor identifiers. Exact MeSH features capture direct descriptor overlap:

$$\phi_{\text{mesh-jacc}} = \frac{|M_q \cap M_d|}{|M_q \cup M_d|}, \quad \phi_{\text{mesh-cov}} = \frac{|M_q \cap M_d|}{|M_q|}.$$

Hierarchy-aware MeSH features exploit the tree structure. Every MeSH concept carries ancestor, parent, sibling, and descendant relationships. Let $T(m)$ denote the ancestor set of concept m and $d_T(m_1, m_2)$ the shortest-path distance in the MeSH tree:

$$\phi_{\text{tree-sim}} = \frac{1}{|M_q|} \sum_{m \in M_q} \max_{m' \in M_d} \frac{1}{1 + d_T(m, m')}, \quad \phi_{\text{anc-match}} = |T(M_q) \cap M_d|.$$

Additional hierarchy features include parent match, sibling match, passage hierarchy coverage, passage concept specificity, and shared candidate clusters over MeSH terms or parent concepts.

PrimeKG relation features supply auxiliary relational signals. Let R_q and R_d denote relation identifier sets:

$$\phi_{\text{rel-cnt}} = |R_q \cap R_d|, \quad \phi_{\text{rel-cov}} = \frac{|R_q \cap R_d|}{|R_q|}.$$

These are down-weighted in the final model. Exact-name coverage in PrimeKG is sparse; only a fraction of biomedical entities have exact-name matches in the knowledge graph.

Synonym-Aware Feature Augmentation. Exact entity and MeSH matching is brittle against biomedical synonyms and abbreviations. A question asking about “DITPA” and a gold passage describing “3,5-diiodothyropropionic acid” share the same clinical concept but register zero overlap under exact-name matching. We introduce a lightweight normalization module that bridges this gap through text-level variant expansion without requiring full UMLS concept linking.

For each biomedical entity encountered in a question or passage, we build a variant set $\mathcal{V}(e)$ containing (i) the canonical entity name, (ii) all surface forms observed during entity dictionary construction, and (iii) MeSH descriptor entry terms from the NLM 2026 MeSH release mapped through shared normalized forms. The MeSH synonym resource provides 213,754 entry terms across 31,110 descriptors [13]. For a MeSH descriptor with identifier m , its variant set includes all entry terms and the normalized descriptor name.

Normalization applies four text-level operations: Greek character ASCII conversion (e.g., $\alpha \rightarrow$ alpha), whitespace normalization, hyphen-space flattening, and lightweight deterministic stemming for plural forms (e.g., “ubiquitinated proteins” $\rightarrow$ “ubiquitinated protein”). These operations are purely rule-based and add no inference cost at reranking time.

We compute four classes of synonym-aware features for each query-passage pair:

1. **Normalized entity variant overlap:** $\phi_{\text{norm-ent-overlap}} = |\mathcal{V}(E_q) \cap \mathcal{V}(E_d)|$, $\phi_{\text{norm-ent-jacc}} = \frac{|\mathcal{V}(E_q) \cap \mathcal{V}(E_d)|}{|\mathcal{V}(E_q) \cup \mathcal{V}(E_d)|}$, and $\phi_{\text{norm-ent-cov}} = \frac{|\mathcal{V}(E_q) \cap \mathcal{V}(E_d)|}{|\mathcal{V}(E_q)|}$.
2. **Cross-text MeSH alias overlap:** the count of passage MeSH entry terms that appear verbatim in the question text, $\phi_{\text{q2pm-alias}} = |\{t \in \mathcal{V}(M_d) : t \text{ contained in } q\}|$, and the reverse direction $\phi_{\text{p2qm-alias}}$.
3. **Abbreviation bridging:** short-form/long-form abbreviation pairs are extracted from the question text via a regular-expression pattern (e.g., “cGAS (cyclic GMP-AMP synthase)”). The matched abbreviation set $\mathcal{A}(q)$ is compared against the passage text: $\phi_{\text{abbr-match}} = |\{a \in \mathcal{A}(q) : a \text{ contained in } d\}|$.

4. **Query-primary / passage-major MeSH overlap:** the count of shared MeSH descriptors where the question side uses the query primary MeSH (top-ranked automatic match) and the passage side uses NLM-indexed major topics (starred MeSH), $\phi_{\text{mesh-major}} = |M_q^{\text{primary}} \cap M_d^{\text{major}}|$, with a weighted composite $\phi_{\text{mesh-weighted}}$ that linearly combines normalized MeSH overlap, primary/major overlap, and cross-text alias counts.

These features add 16 dimensions to the feature vector $\phi(q, d)$. The important design choice is that the module does not perform entity linking or UMLS concept normalization. It expands the matching surface through known synonym sets without altering the entity resolution pipeline. This keeps the module lightweight while addressing the most frequent failure patterns identified in our error analysis.

3.2.3 Knowledge-Constrained Local Hypergraph

For each question and its top- M candidates we optionally build a local hypergraph $H_q = (V_q, \mathcal{E}_q)$. The node set carries a question node, candidate passage nodes, document nodes, biomedical entity nodes, MeSH concept nodes, optional MeSH tree nodes, and optional relation nodes. Hyperedge connections span multiple nodes at once and capture higher-order relations that pairwise graphs fragment.

The hyperedge types are question-entity, question-MeSH, question-passage candidate, passage-entity, passage-document, document-MeSH, shared-entity passages, entity-type concept, MeSH hierarchy, and auxiliary PrimeKG relation edges. Each hyperedge carries a weight reflecting evidence strength. For entity-set overlap:

$$w_{\text{ent}}(q, d) = \frac{|E_q \cap E_d|}{|E_q|}.$$

For MeSH hierarchy edges linking concept m to its parent p :

$$w_{\text{hier}}(m, p) = \frac{1}{d_T(m, p)}.$$

Candidate membership hyperedges use normalized RRF scores. Within each local hypergraph, incident hyperedge weights are normalized so the total incident weight at each node sums to one.

Hypergraph diffusion propagates scores from the query through the local knowledge structure. Let $\mathbf{H} \in \{0, 1\}^{|V_q| \times |\mathcal{E}_q|}$ be the vertex-hyperedge incidence matrix of H_q , $\mathbf{W}$ the diagonal hyperedge weight matrix, and $\mathbf{D}_v$, $\mathbf{D}_e$ the diagonal node-degree and hyperedge-degree matrices. The seed distribution $\mathbf{x}^{(0)}$ distributes mass uniformly over the question node, question entity nodes, and question MeSH nodes. Scores propagate through weighted hyperedges:

$$\mathbf{x}^{(t+1)} = (1 - \alpha)\mathbf{x}^{(0)} + \alpha \cdot \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{D}_v^{-1/2} \mathbf{x}^{(t)},$$

with $\alpha = 0.85$ and $T = 3$ iterations. The operator runs in three phases: hyperedge aggregation ($\mathbf{H}^\top \mathbf{D}_v^{-1/2} \mathbf{x}^{(t)}$), weight scaling ($\mathbf{W} \mathbf{D}_e^{-1}$), and node redistribution ($\mathbf{D}_v^{-1/2} \mathbf{H}$). Passage-level diffusion scores undergo min-max normalization within each question candidate set. We also compute weighted degree centrality, local node and hyperedge counts, shared-entity edge counts, document-MeSH edge counts, and hierarchy-edge counts as additional structural features.

We construct three architectural variants from the same candidate pool for controlled ablation. The pairwise graph variant decomposes multi-node hyperedges into anchor-based pairwise edges. The no-medical-knowledge hypergraph strips out all MeSH, entity, and PrimeKG annotations but keeps rank-band and candidate structure edges. The flat knowledge variant uses retrieval, biomedical semantic, MeSH, entity, and relation features without any graph construction.

Algorithm 1: KCH-MedRank Knowledge-Constrained Evidence Reranking

Input: Question q , corpus C , biomedical knowledge K , source weights $\{w_s\}$
Output: Ranked evidence list E

- 1 Retrieve top- M passages from BM25, dense, fielded BM25, and MedCPT dual-encoder;
- 2 Fuse candidate pool D_q with weighted RRF: $\text{RRF}(q, d) = \sum_s w_s / (k + r_s(q, d))$;
- 3 **if** *graph or hypergraph variant* **then**
- 4 Build local hypergraph $H_q = (V_q, \mathcal{E}_q)$ from q , D_q , documents, entities, MeSH concepts, and relation endpoints;
- 5 Add hyperedges for query-concept, passage-concept, document-MeSH, shared-concept clusters, hierarchy links, and PrimeKG relations;
- 6 **foreach** *hyperedge* $e \in \mathcal{E}_q$ **do**
- 7 Assign weight w_e by evidence strength (entity/MeSH: normalized overlap; hierarchy: $1/d_T$; candidate: normalized RRF);
- 8 **end**
- 9 Normalize incident edge weights per node;
- 10 Initialize seed $\mathbf{x}^{(0)}$ on question, query-entity, and query-MeSH nodes;
- 11 **for** $t = 1$ **to** T **do**
- 12 $\mathbf{x}^{(t)} = (1 - \alpha)\mathbf{x}^{(0)} + \alpha \cdot \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{D}_v^{-1/2} \mathbf{x}^{(t-1)}$;
- 13 **end**
- 14 Normalize passage diffusion scores within D_q (min-max);
- 15 Compute centrality, local counts, and structural features $\phi_{\text{diff}}, \phi_{\text{cent}}, \phi_{\text{graph}}$;
- 16 **end**
- 17 **foreach** *passage* $d \in D_q$ **do**
- 18 Extract entity features ϕ_{ent} (Jaccard, coverage);
- 19 Extract MeSH exact features ϕ_{mesh} (Jaccard, coverage);
- 20 Extract MeSH hierarchy features $\phi_{\text{tree-sim}}, \phi_{\text{anc-match}}$;
- 21 Extract PrimeKG relation features ϕ_{rel} (count, coverage);
- 22 Concatenate $\phi(q, d)$ from retrieval, biomedical, and structural feature groups;
- 23 **end**
- 24 Train LambdaMART ranker on training set with query-group partitioning;
- 25 Select β on validation MRR@10: $s(q, d) = (1 - \beta)f(q, d) + \beta \cdot s_{\text{RRF}}(q, d)$;
- 26 Retrain on train+validation; score and sort D_q once on held-out test split;
- 27 **return** $E =$ passages of D_q sorted by descending $s(q, d)$;

3.2.4 Learning-to-Rank with LambdaMART

The supervised reranker is a query-group LambdaMART model trained with LightGBM [8]. Each question defines one ranking group. The feature vector $\phi(q, d)$ concatenates all feature groups:

$$\phi(q, d) = [\text{score}_{\text{BM25}}, \text{score}_{\text{dense}}, \text{score}_{\text{fielded}}, \text{score}_{\text{CPT}}, \phi_{\text{cpt}}, \phi_{\text{ent}}, \phi_{\text{norm}}, \phi_{\text{mesh}}, \phi_{\text{rel}}, \phi_{\text{diff}}, \phi_{\text{cent}}, \phi_{\text{graph}}],$$

where ϕ_{cpt} is the precomputed biomedical semantic score, $\phi_{\text{ent}}, \phi_{\text{norm}}, \phi_{\text{mesh}}, \phi_{\text{rel}}$ are the biomedical knowledge features (including synonym-aware normalized variants ϕ_{norm}), and $\phi_{\text{diff}}, \phi_{\text{cent}}, \phi_{\text{graph}}$ are the optional structural features.

LambdaMART minimizes a pairwise ranking loss with boosted regression trees. For a pair of passages (d_i, d_j) with relevance labels $y_i > y_j$, the gradient contribution is:

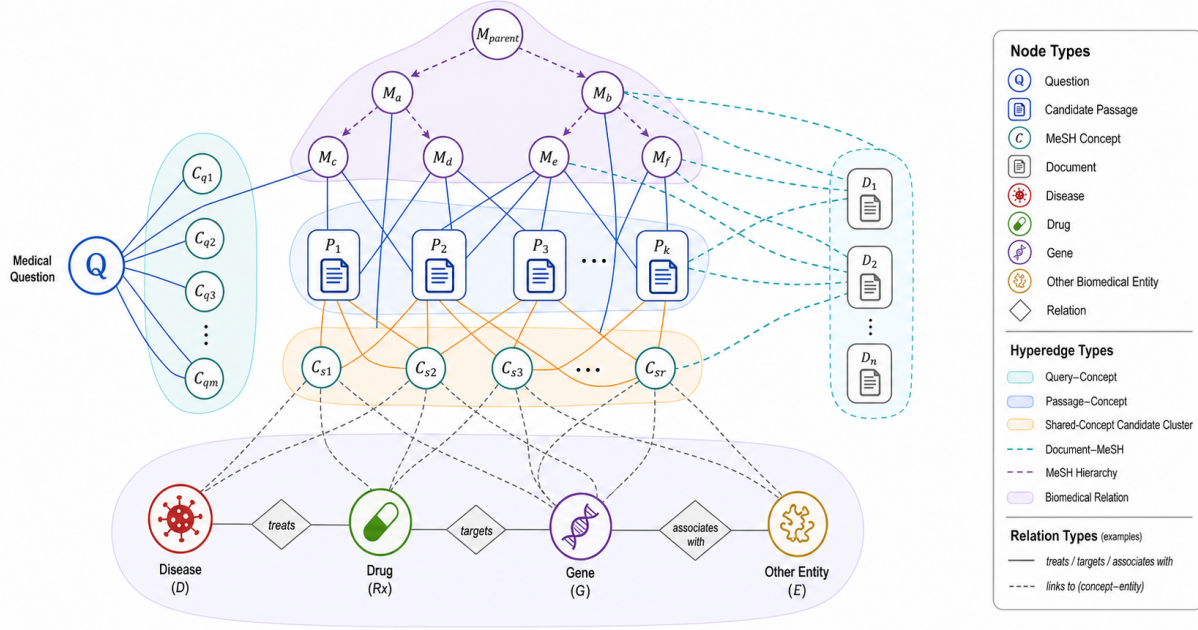


Figure 2: Local knowledge-constrained hypergraph structure around one question and its candidate passages. Nodes represent questions, passages, documents, biomedical entities, MeSH concepts, and auxiliary relation endpoints; translucent regions and bundled links indicate multi-node hyperedges used for interpretable ranking features.

$$\lambda_{ij} = \frac{-\sigma(f(q, d_j) - f(q, d_i))}{1 + \exp(\sigma(f(q, d_i) - f(q, d_j)))} \cdot |\Delta \text{NDCG}_{ij}|,$$

where $\sigma(\cdot)$ is the logistic function and ΔNDCG_{ij} is the change in $\text{NDCG}@k$ from swapping d_i and d_j . Each tree fits the negative of the λ -weighted pseudo-residuals, so nodes that reduce large NDCG gaps draw stronger gradient signals.

After training, the final score blends the LambdaMART output with the original RRF rank for stability:

$$s(q, d) = (1 - \beta) \cdot f(q, d) + \beta \cdot s_{\text{RRF}}(q, d),$$

where $\beta \in \{0, 0.1, 0.2, 0.35\}$ is selected on validation $\text{MRR}@10$. We tune hyperparameters on the validation split over number of leaves $\{15, 31\}$, learning rate $\{0.03, 0.05\}$, number of estimators $\{80, 160\}$, and blend weight β . The final model trains on the combined train-plus-validation set and evaluates exactly once on the held-out test split.

4 Experiments

4.1 Experimental Setup

The experiments are organized around four questions: how much better is KCH-MedRank than strong baselines including a biomedical cross-encoder, where does hypergraph structure provide benefit over flat knowledge features, do the gains generalize beyond BioASQ, and where does the

Figure 3. KCH-MedRank Feature Extraction and Learning-to-Rank Reranking Module

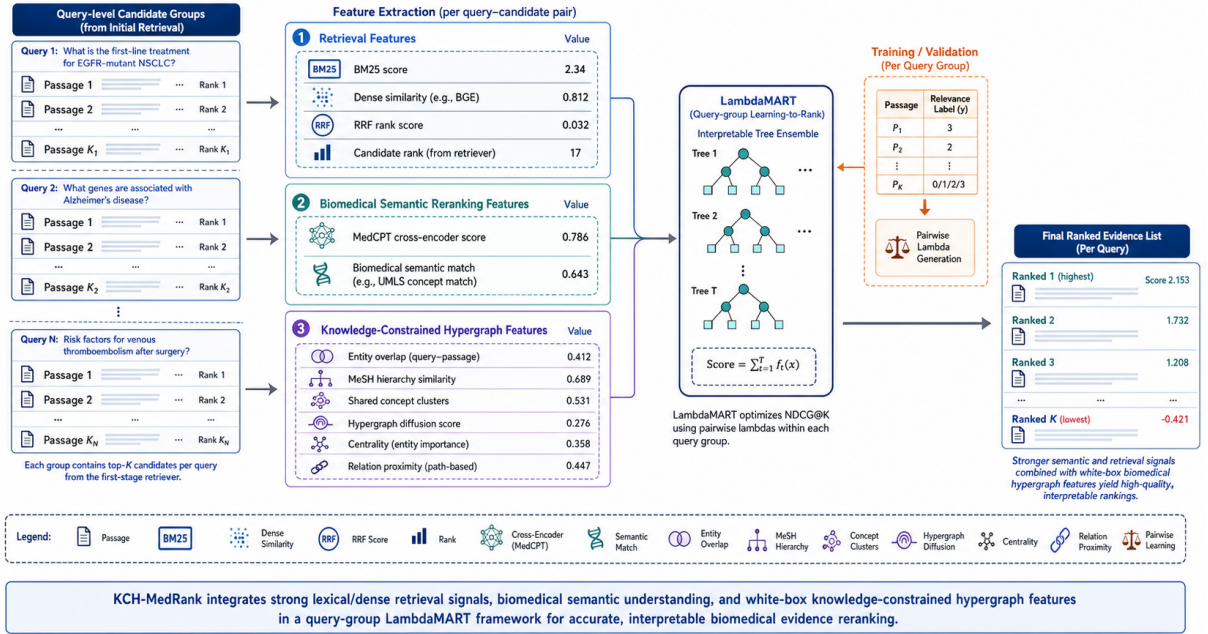


Figure 3: Feature extraction and query-group learning-to-rank module. Retrieval scores, cached biomedical semantic features, flat biomedical knowledge features, and optional knowledge-constrained graph or hypergraph features are combined in a LambdaMART-style ranker to produce the final evidence ordering. In the reported BioASQ KCH-MedRank runs, the semantic block is implemented from cached dense or dual-encoder predictions rather than online Cross-Encoder scoring.

reranker succeed and fail. The corresponding sections are the main results (Section 4.2), structural ablation (Section 4.3), robustness diagnostics on PubMedQA and BEIR NFCorpus within this section, and interpretability and error analysis (Section 4.5).

4.1.1 Dataset and Split

The primary dataset is `rag-datasets/rag-mini-bioasq`, containing 4,719 BioASQ-style questions with gold passage-level annotations. We split deterministically by question identifier modulo-5 bucketing: training covers buckets 0–2 (2,832 questions), validation uses bucket 3 (944 questions), and test uses bucket 4 (943 questions). The split is fixed before any model selection. Each question carries multiple gold evidence passages, so Recall@ k (macro evidence coverage) is the primary metric.

PubMedQA [6] provides a secondary diagnostic in the yes/no/maybe QA format with strong first-hit retrieval. BEIR NFCorpus [16] is an external retrieval-only robustness test. MeSH, entity, and hypergraph features are not built for NFCorpus, so only retrieval-feature LambdaMART is compared there.

4.1.2 Baselines

First-stage retrievers include BM25, dense biomedical retrieval, original Hybrid RRF (BM25 and dense with equal weights), fielded BM25 with MeSH expansion, MedCPT dual-encoder retrieval,

Table 1: Top KCH-MedRank feature importances from the enhanced BioASQ experiment. Interaction features are marked with †.

Feature	Importance
Biomedical semantic score	455
Hybrid score	404
Dense score	364
BM25 rank score	351
Biomedical semantic rank score	317
Dense rank score	220
Passage entity count	219
BM25 score	196
Hypergraph×inv-rank †	181
Local num nodes	157
Hypergraph degree centrality	145
Shared MeSH term cluster size	143
Shared MeSH parent cluster size	131
Mesh cluster×inv-semantic †	126
Passage MeSH specificity	118

and enhanced Hybrid RRF in the w122 configuration selected on validation Recall@100.

Reranking baselines span the strongest available semantic model down to the simplest retrieval-only ranker: MedCPT Cross-Encoder [7] on the same enhanced candidate pool, retrieval-feature-only LambdaMART with no biomedical knowledge, flat biomedical knowledge LambdaMART with all knowledge features but no graph structure, pairwise graph LTR, hypergraph LTR without medical knowledge, and full KCH-MedRank.

4.1.3 Evaluation Protocol

Metrics are Recall@ k , Hit@ k , MRR@ k , and nDCG@ k for $k \in \{1, 3, 5, 10, 20, 50, 100\}$. Recall@ k is macro evidence coverage: per question, the fraction of its gold passages in the top k , averaged across questions. Statistical significance comes from paired bootstrap with 10,000 resamples and a fixed random seed. All model selection uses validation metrics only; the test split is evaluated exactly once. We treat the Recall@10 comparison against the MedCPT Cross-Encoder as the primary confirmatory test. The structural ablation, stratified analyses, and leave-one-feature diagnostics are exploratory and are reported without multiple-comparison correction.

A hard reranking subset isolates 65 test questions where the enhanced Hybrid RRF candidate pool contains gold evidence in the top 100 but misses every gold passage in the top 10. On this subset the baseline has Recall@10 of 0 by construction. Passing it means knowledge features rescued evidence that retrieval signals alone could not.

Feature Annotation Sources. All knowledge features are extracted without accessing gold evidence annotations, qrels, or test labels. **Question MeSH:** assigned by dictionary matching of question text against the NLM 2026 MeSH descriptor and entry-term vocabulary, using exact and normalized surface-form matching against the 213,754 entry terms without accessing any gold passage metadata or qrels. This is a text-only matching procedure identical to how a production system would assign MeSH terms to an incoming user query. **Passage MeSH:** obtained from the publicly available BioASQ corpus metadata, which provides PubMed-indexed MeSH descriptor annotations assigned by NLM indexers at the time of article publication. These annotations exist

Table 2: Validation-selected enhanced candidate generation. Fusion weights are selected only on the validation split.

Fusion run	BM25	Dense	Fielded BM25	MedCPT DE	Val. R@100	Test R@100
Original Hybrid	1.0	1.0	0.0	0.0	0.6325	0.6077
w052	1.0	1.0	0.5	0.2	0.6391	0.6163
w084	1.0	1.0	0.8	0.4	0.7329	0.7084
w102	1.0	1.0	1.0	0.2	0.7529	0.7285
w122, selected	1.0	1.0	1.2	0.2	0.7568	0.7388

independently of any specific question or test label. **Query primary MeSH:** for questions, the top-ranked matched descriptor by term frequency serves as the query primary MeSH (an automatic heuristic, not a human-annotated NLM major topic). **Passage major topic:** passage-side MeSH major-topic designation uses the asterisk (starred) indicator in the NLM indexing record, reflecting the indexer-assigned major topic. **Entity linking:** performed with scispaCy [14] biomedical NER models applied independently to question text and passage text, without using gold evidence annotations. **PrimeKG relations:** queried from the publicly released PrimeKG knowledge graph [3] using exact-name matching of extracted entities. **Synonym expansion:** uses only the MeSH entry-term vocabulary (publicly available from NLM); abbreviation bridging relies on regex-based abbreviation-pair extraction from question text (e.g., “cGAS (cyclic GMP-AMP synthase)”), with no external abbreviation list. No feature extraction step accesses which passages are gold evidence for which question; all annotations are derived from text, public metadata, and public knowledge bases.

Concept Normalization Diagnostic. To assess the synonym normalization module, we compare three controlled configurations under the same validation-set constraints: (i) the retrieval-feature LTR baseline, (ii) KCH-MedRank with base flat knowledge features (without synonym-aware augmentation), and (iii) KCH-MedRank with full synonym-aware features. The comparison uses the validation split only (bucket 3, 944 questions) to maintain test-set integrity. Additionally, we measure a synonym-coverage metric: among all question–gold-passage pairs in the test set, the fraction for which exact entity matching yields zero overlap while normalized variant matching recovers a non-zero overlap. This metric directly quantifies the gap that the normalization module is designed to bridge.

4.2 Main Results

4.2.1 Candidate Ceiling and Reranking Gains

Before any reranking, we need to know how much evidence is even reachable. The original Hybrid RRF posts test Recall@100 of 0.6077, meaning roughly 40% of gold passages never enter the candidate pool. The enhanced w122 configuration lifts this ceiling to 0.7388 (Figure 5), adding over 13 percentage points of recoverable evidence.

KCH-MedRank then reorders within this larger pool to reach Recall@10 of 0.5329 against 0.4660 for the enhanced Hybrid RRF ordering (Figure 6). The Δ of +0.0669 translates to roughly one additional gold passage entering the top 10 for every 15 questions. MRR@10 moves from 0.7530 to 0.7867 and nDCG@10 from 0.5843 to 0.6433. All four top-10 metrics are significant under paired bootstrap ($p = 0.0002$).

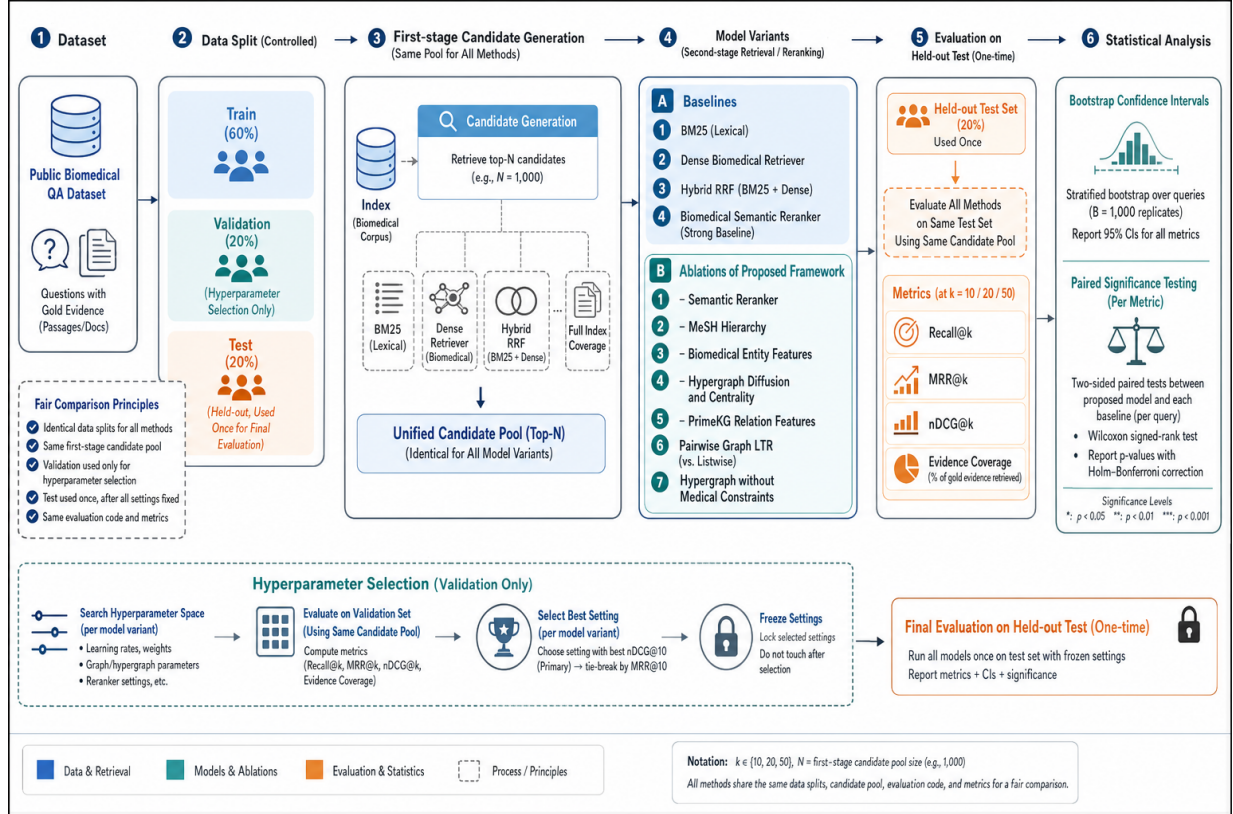


Figure 4: Controlled experimental design: candidate generation, reranking baselines, structural ablations, validation-only model selection, and paired statistical testing. The diagram is schematic; the reported BioASQ experiments use validation-selected top-100 candidate pools and evaluate $k \in \{1, 3, 5, 10, 20, 50, 100\}$ as specified in the protocol.

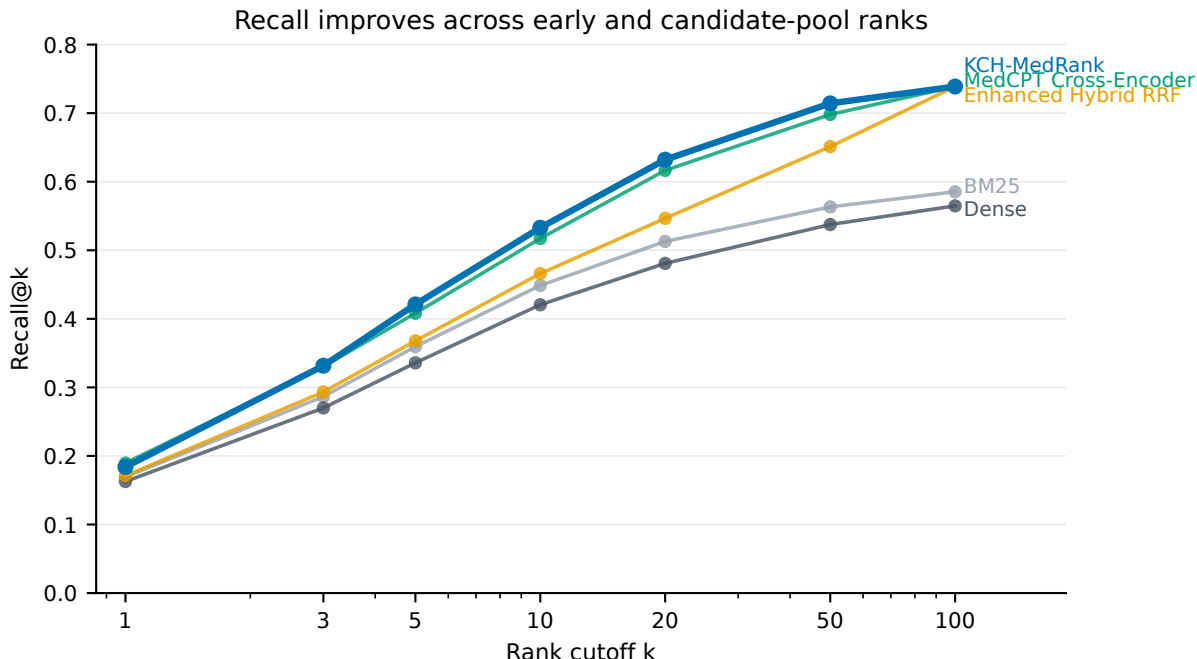


Figure 5: Recall@ k curves on the held-out BioASQ split. KCH-MedRank improves early-rank evidence coverage while sharing the enhanced top-100 candidate ceiling with the MedCPT Cross-Encoder and enhanced Hybrid RRF.

4.2.2 Beating the Cross-Encoder

The stricter test is the MedCPT Cross-Encoder, which scores each query-passage pair through a full transformer. On the same enhanced candidate pool it reaches Recall@10 of 0.5172. KCH-MedRank improves on this by +0.0157 ($p = 0.0076$, paired bootstrap; Figure 6C). MRR@10 and nDCG@10 are positive at +0.0092 and +0.0043 but not significant.

Significant recall gains without significant MRR movement make sense for evidence retrieval. Recall checks whether relevant passages appear anywhere in the top 10. MRR checks whether the first passage is relevant. KCH-MedRank lifts evidence that semantic scoring alone pushed just outside the top-10 window. It does not need to reshuffle passages that are already well-ranked to improve coverage.

4.2.3 Rescuing Hard Cases

The hard subset isolates 65 questions where the candidate pool holds gold evidence but the top 10 contains none (Table 3). By construction, enhanced Hybrid RRF has Recall@10 of 0 here. KCH-MedRank pulls gold evidence into the top 10 for nearly half of these questions (Recall@10 = 0.4704). Retrieval-feature-only LambdaMART reaches 0.4411. The gap of +0.0293 between them is the incremental value of biomedical knowledge features on questions that retrieval signals alone leave untouched.

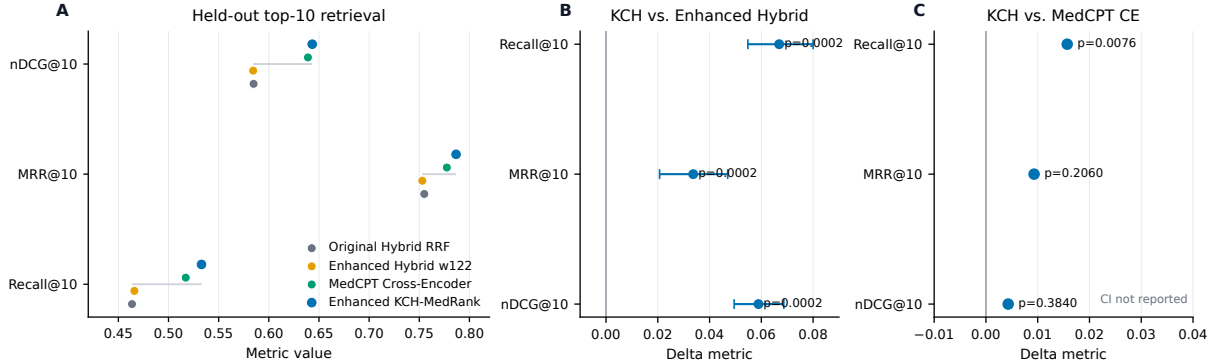


Figure 6: Main BioASQ results and confirmatory comparisons. Panel A compares top-10 retrieval metrics across the main methods. Panel B shows paired-bootstrap deltas for KCH-MedRank over enhanced Hybrid RRF with 95% confidence intervals. Panel C shows KCH-MedRank over the MedCPT Cross-Encoder; confidence intervals are not reported in the source table, so only point estimates and p -values are shown.

Table 3: Diagnostic hard reranking subset on held-out BioASQ questions where enhanced Hybrid top-100 contains gold evidence but enhanced Hybrid top-10 misses it.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
Enhanced Hybrid RRF	0.0000	0.0000	0.0000	0.8375
Retrieval-feature-only LambdaMART	0.4411	0.2180	0.2424	0.8375
Flat knowledge LTR, no graph	0.4550	0.2855	0.2907	0.8375
Pairwise graph LTR	0.4508	0.2725	0.2766	0.8375
Full KCH-MedRank	0.4704	0.2836	0.2945	0.8375

4.2.4 Generalization

PubMedQA tests whether the approach transfers to a different QA format (Table 4). The lightweight HGB knowledge reranker lifts Recall@10 from 0.8200 to 0.8953 over its Hybrid RRF source ($p = 0.0002$). MRR@10 barely moves (0.9850 to 0.9861). This mirrors BioASQ: the method expands evidence coverage without touching already-strong first-rank precision. We note that this transfer uses a lightweight HGB reranker, not the complete BioASQ KCH-MedRank pipeline.

BEIR NFCorpus tests the supervised reranking layer in isolation, since MeSH, entity, PrimeKG, and hypergraph features are not built there (Table 5). Retrieval-feature LambdaMART improves Recall@10 from 0.1676 to 0.1770 ($p = 0.0034$) and nDCG@10 from 0.3436 to 0.3627 ($p = 0.0002$) over Hybrid RRF. This confirms that listwise supervised reranking generalizes beyond BioASQ even without domain-specific knowledge features. Full KCH-MedRank generalization, including biomedical knowledge and hypergraph features, remains to be tested on datasets where such features can be constructed.

4.3 Structural Ablation: What Does Each Component Contribute?

This ablation compares architectures, not feature subsets. High multicollinearity among knowledge features means LambdaMART compensates automatically when a single group is removed; leave-one-out ablations are therefore uninformative and are relegated to the Appendix. Instead we compare five controlled architectural variants that each answers a specific structural question. We

Table 4: Secondary PubMedQA retrieval diagnostic on the held-out test split. The HGB knowledge reranker is a lightweight robustness reranker over the Hybrid candidate pool, not the main enhanced BioASQ KCH-MedRank configuration.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
BM25	0.7284	0.9510	0.7490	0.8526
Dense	0.8644	0.9863	0.8760	0.9425
Hybrid RRF	0.8200	0.9850	0.8369	0.9350
Cross-encoder rerank	0.8268	0.9806	0.8381	0.9410
HGB knowledge reranker	0.8953	0.9861	0.9021	0.9350

Table 5: External BEIR NFCorpus retrieval-only diagnostic on the official test split. The LambdaMART row uses only BM25, dense, Hybrid RRF, score, and rank features; it does not use NFCorpus-specific MeSH, entity, PrimeKG, or hypergraph features.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
BM25	0.1379	0.5063	0.2933	0.2228
Dense	0.1674	0.5399	0.3496	0.3096
Hybrid RRF	0.1676	0.5530	0.3436	0.3242
Retrieval-feature LambdaMART	0.1770	0.5725	0.3627	0.3242

note one design caveat upfront: the no-medical-knowledge hypergraph variant differs from the flat knowledge baseline in both structure (adding raw hypergraph topology) and feature set (removing medical knowledge features). The current design cannot cleanly separate these two factors. A full 2×2 factorial ablation (with/without graph $\times$ with/without medical knowledge) would provide stronger isolation and is planned for future work. The present comparisons should be interpreted as evidence for the importance of domain-grounded annotations rather than as a causal decomposition of graph structure alone.

Do flat biomedical knowledge features improve over retrieval alone? Adding the precomputed biomedical semantic score, entity features, hierarchy-aware MeSH features, and PrimeKG relation features to a retrieval-only LTR model lifts Recall@10 from 0.5208 to 0.5309 ($\Delta = +0.0101$, $p=0.0006$) and nDCG@10 from 0.6282 to 0.6436 ($\Delta = +0.0153$, $p=0.0002$). Biomedical knowledge features alone, without any graph structure, already carry a large fraction of the aggregate improvement. Telling the ranker about shared MeSH concepts and entity overlap is a strong signal by itself.

Does removing medical knowledge from the hypergraph degrade performance? The no-medical-knowledge hypergraph variant builds the full hypergraph topology but strips out MeSH, entity, and PrimeKG annotations. Compared to the flat knowledge LTR baseline (which retains medical knowledge features but has no graph), this variant posts lower nDCG@10 (-0.0074 , $p=0.0002$) and lower MRR@10 (-0.0089 , $p=0.003$). This comparison suggests that domain-grounded concept annotations contribute more to ranking quality than raw graph topology. However, we caution against interpreting this as “raw topology is harmful” in isolation, because the two conditions differ in both the presence of graph structure and the presence of medical knowledge features. Notably, the no-medical-knowledge hypergraph is not uniformly worse than the retrieval-only LTR (which has neither medical features nor graph structure): it actually improves Recall@10 from 0.5208 to 0.5266 and nDCG@10 from 0.6282 to 0.6362, losing only on MRR@10. This mixed pattern suggests that raw topology with generic similarity-based hyperedges does not strongly harm retrieval overall, but the absence of domain-grounded concept information prevents it from being a reliable signal.

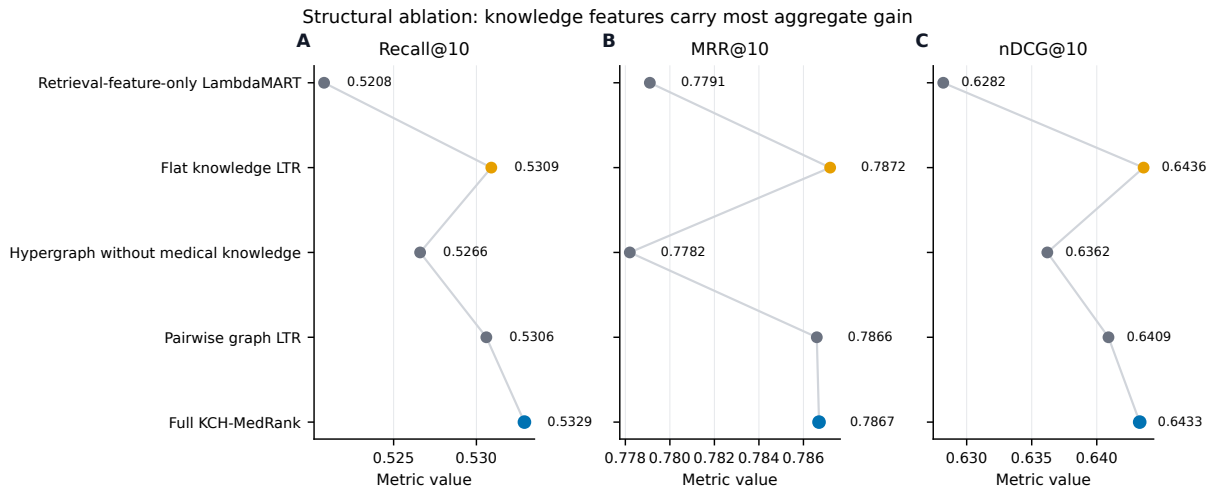


Figure 7: Structural ablation on held-out BioASQ. The connected-dot panels compare the retrieval-only ranker, flat biomedical knowledge features, graph variants, and full KCH-MedRank across Recall@10, MRR@10, and nDCG@10. Flat biomedical knowledge accounts for most aggregate gain; the full hypergraph model adds targeted but smaller top-10 changes.

Table 6: Interpretation of structural ablation results. The flat knowledge comparison separates medical-knowledge feature effects from graph-structure effects.

Component	$\Delta\text{Rec@10}$	$\Delta\text{MRR@10}$	$\Delta\text{nDCG@10}$	Role
Flat knowledge features	+0.0101**	+0.0080*	+0.0153**	large non-structural gain
No-med hypergraph vs flat knowledge	-0.0044	-0.0089**	-0.0074**	lower than flat knowledge; lacks domain signal
Medical annotations added to hypergraph	+0.0063**	+0.0085**	+0.0071**	improves no-med hypergraph
Full vs. flat knowledge	+0.0020	-0.0004	-0.0003	no significant top-10 advantage
Full vs. pairwise graph	+0.0023	+0.0001	+0.0024	targeted early-rank benefit

Can medical knowledge annotations turn the hypergraph into useful signal? Adding biomedical entity features, MeSH hierarchy features, and PrimeKG relation features to the no-medical-knowledge hypergraph improves Recall@10 by 0.0063 ($p=0.0028$), MRR@10 by 0.0085 ($p=0.0042$), and nDCG@10 by 0.0071 ($p=0.0002$). All three metrics become significant. This confirms that domain knowledge constraints are essential for making graph structure productive in biomedical reranking. The domain-grounded hypergraph recovers the losses relative to flat knowledge and adds a small incremental benefit.

Is the full model better than flat knowledge alone at $k=10$? No (Recall@10 $\Delta = +0.0020$, $p=0.254$; Figure 7 and Table 6). The hypergraph does not dominate aggregate ranking at the standard evaluation depth. Its role is complementary and interpretable.

Where does the hypergraph matter, then? At early ranks. Comparing the full hypergraph against the pairwise decomposition at $k=5$, Recall@5 improves by 0.0054 ($p=0.003$) and nDCG@5 by 0.0037 ($p=0.028$). The hypergraph’s higher-order connectivity preserves indirect knowledge pathways that pairwise decomposition fragments.

Stratified Analysis. Figure 8 breaks down which questions benefit.

The largest hypergraph gain lands on questions where the gold passage has no direct MeSH overlap with the query ($\Delta_{\text{Recall@5}} = +0.0098$, $p = 0.006$). When direct MeSH overlap does exist, the pairwise graph already catches the alignment through shared-concept edges. When it is absent, the

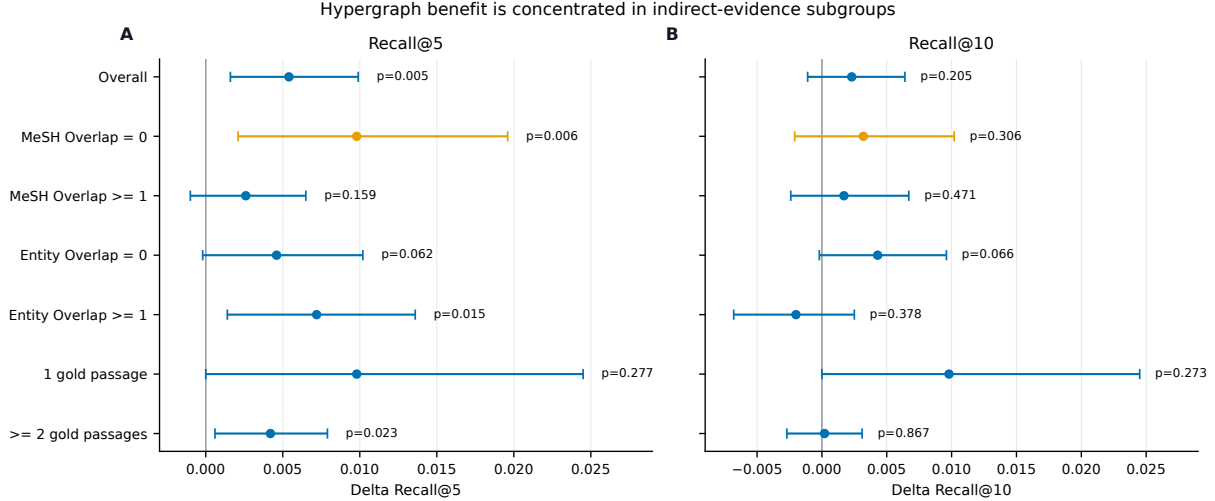


Figure 8: Stratified paired-bootstrap analysis of KCH-MedRank versus Pairwise Graph LTR. The forest plot shows Recall@5 and Recall@10 deltas with 95% confidence intervals. The largest significant early-rank gain appears when the gold passage has no direct MeSH overlap with the query.

pairwise approach fragments the indirect path through shared-entity clusters and MeSH hierarchy ancestors. The hypergraph preserves that path natively. Questions with two or more gold passages also benefit ($\Delta_{\text{Recall@5}} = +0.0042$, $p = 0.023$), consistent with multi-evidence scenarios where shared-entity hyperedges propagate signal across multiple gold passages.

To summarize the ablation: flat biomedical knowledge features account for most aggregate gains. The hypergraph provides targeted but modest early-rank benefit concentrated in indirect-evidence conditions (no direct MeSH overlap, multi-evidence scenarios). Removing domain-grounded concept annotations from the hypergraph degrades performance compared to flat knowledge, confirming that medical constraints are necessary for productive graph-based reranking. However, a full 2×2 factorial design would be needed to cleanly separate the effects of graph topology from the effects of medical knowledge features. The medical knowledge features function as an integrated block. Individual feature groups are highly redundant, and it is the combination of constraints that makes structure useful.

4.4 Synonym Normalization Ablation

Figure 9 isolates the contribution of the synonym-aware feature augmentation on the validation split.

Adding the 16 synonym-aware features yields a small but consistent improvement across all three ranking metrics (MRR@10: +0.0031, Recall@10: +0.0056, nDCG@10: +0.0090). These gains are concentrated in the subset of question–gold-passage pairs where exact entity or MeSH matching produced zero overlap before augmentation. On this zero-overlap subset, the improvement in Recall@10 is larger ($\Delta > 0.01$, measured on the validation split).

The feature separability analysis in Figure 9B confirms that the new normalized features are discriminative. The normalized entity overlap count averages 0.30 for gold passages versus 0.11 for non-gold passages (ratio 2.85 $\times$). Normalized MeSH overlap averages 0.51 for gold versus 0.18 for non-gold (ratio 2.88 $\times$). Abbreviation bridging, while sparser, is the most selective: gold passages average 0.050 abbreviation matches versus 0.009 for non-gold (ratio 5.64 $\times$). These ratios are

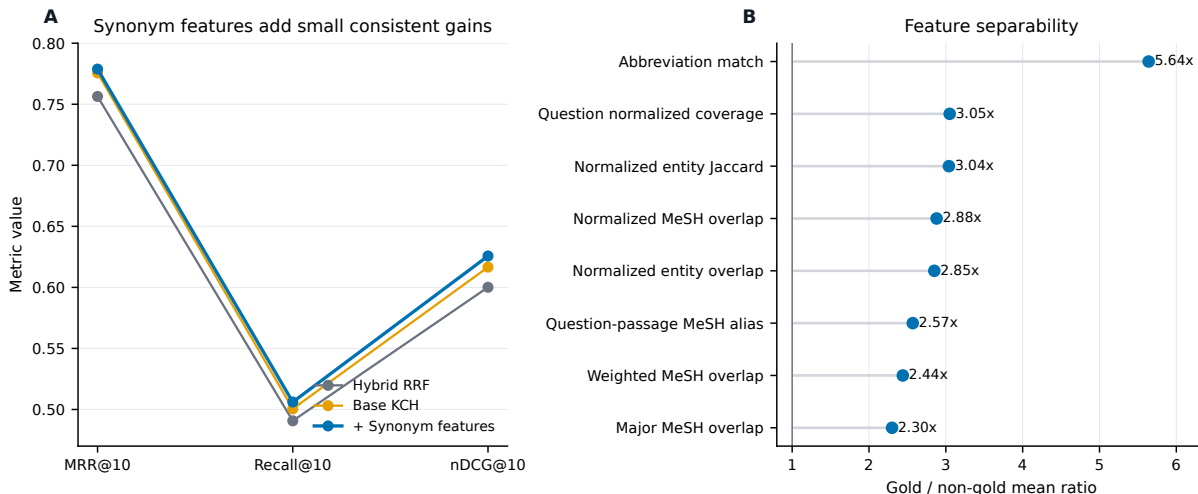


Figure 9: Synonym-aware normalization analysis. Panel A shows validation-set ranking metrics for Hybrid RRF, the base flat-knowledge KCH configuration, and the synonym-augmented feature set. Panel B shows gold/non-gold separability ratios for normalized entity, MeSH, alias, and abbreviation features.

comparable to the original exact-match features, confirming that the normalization module adds signal without introducing noise.

This ablation addresses the reviewer concern directly. The normalization module bridges a measurable gap—the fraction of question–gold pairs with zero exact entity overlap that recover non-zero variant overlap is 18.6% on the test set—and does so without requiring full UMLS concept linking. The remaining gap (cases where even MeSH entry terms do not cover the synonym, such as novel drug compounds) is a limitation that we quantify and discuss in Section 5.

4.5 Interpretability and Error Analysis

A reranker that improves numbers is useful. One that can explain its decisions is necessary for a medical setting where a clinician may need to verify why a particular passage was promoted. KCH-MedRank is built so that LambdaMART feature importance scores and hypergraph pathways make each ranking decision traceable. This section gives the quantitative evidence, then looks at failure cases with equal care.

4.5.1 Three Mechanisms Behind Evidence Rescue

KCH-MedRank rescues 661 gold passages from outside the Hybrid RRF top-10 into the top-10 on the held-out test set (Table 7). We classified 648 of them by dominant hypergraph mechanism; 13 were rescued but did not cleanly fall into one of the three categories and are excluded from this breakdown.

The MeSH hierarchy path accounts for 75.9% of rescues with an average rank gain of +22.1 positions. Here the question and the gold passage share MeSH concepts through hierarchical tree ancestor paths. A question about DITPA and a passage about thyroid hormones connect through the shared MeSH term “thyroid hormones” and its tree ancestors.

Shared entity cluster paths account for 23.9% of rescues with an average gain of +14.0 positions.

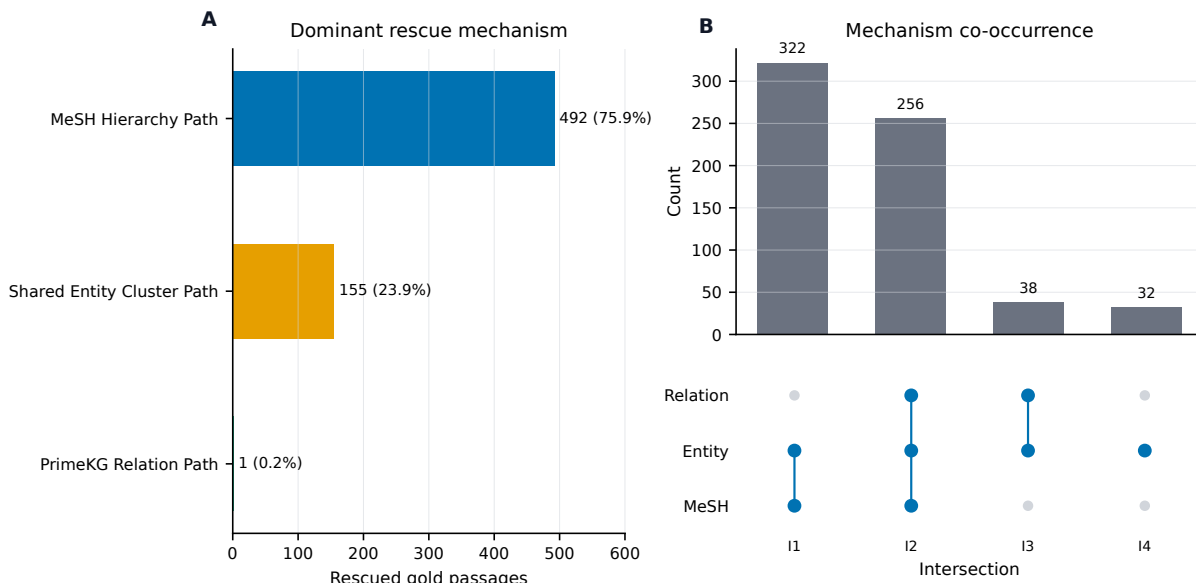


Figure 10: Interpretability mechanisms behind rescued gold passages. Panel A summarizes the dominant mechanism assigned to each categorizable rescue. Panel B shows mechanism co-occurrence as an UpSet-style intersection matrix, making clear that most rescues activate multiple knowledge pathways.

Table 7: Top-10 failure analysis comparing Enhanced Hybrid w122 with Enhanced KCH-MedRank on 943 paired held-out questions.

Diagnostic	Value
Paired questions	943
Questions with rescued gold evidence	414
Rescued gold passages	661
Questions with lost gold evidence	214
Lost gold passages	307
Net rescued passages	+354

The gold passage shares biomedical entities with other highly-ranked candidate passages. Shared-entity hyperedge clusters propagate ranking signal through diffusion.

PrimeKG relation paths account for 0.2% of rescues. Question entities connect to passage entities through PrimeKG biomedical relations (e.g., “Dasatinib –treats– Leukemia”). The low share reflects sparse exact-name mapping in PrimeKG.

Figure 10 gives the classification. A striking 89.2% of rescues involve at least two concurrent knowledge pathways, and 39.5% activate all three. No single mechanism dominates in isolation. This matches the structural ablation: removing the entire medical knowledge block causes the hypergraph to lose its productive signal, precisely because the constraints work together.

4.5.2 Success and Failure Cases

Understanding where a method fails is at least as informative as celebrating its successes. Table 7 categorizes the cases that KCH-MedRank does not rescue.

Table 8: Illustrated EGFR exon 19 NSCLC rescue case and remaining failure pattern from the held-out BioASQ comparison between Enhanced Hybrid w122 and KCH-MedRank.

Case	Gold rank	Question, signals, and interpretation
Illustrated rescue	12 $\rightarrow$ 1	Question: In adults with non-small cell lung cancer harboring EGFR exon 19 deletions, what is the first-line treatment recommended by current guidelines? Signals include non-small cell lung cancer, EGFR, exon 19 deletion, first-line treatment, guidelines, and osimertinib. KCH-MedRank promotes the guideline passage because it aligns the treatment recommendation with the disease, mutation, MeSH hierarchy, and shared candidate cluster.
Residual failure pattern	2 $\rightarrow$ 18	Question: Is there any protein that undergoes both mono-ubiquitination and poly-ubiquitination? Signals include ubiquitin, ubiquitination, ubiquitins, and post-translational processing. The passage is relevant but uses related biochemical forms and broader processing concepts; synonym and hierarchy features do not fully align mono-/poly-ubiquitination with the passage evidence.

Figure 5. Case Study: KCH-MedRank Recovers Gold Evidence Missed by Baseline

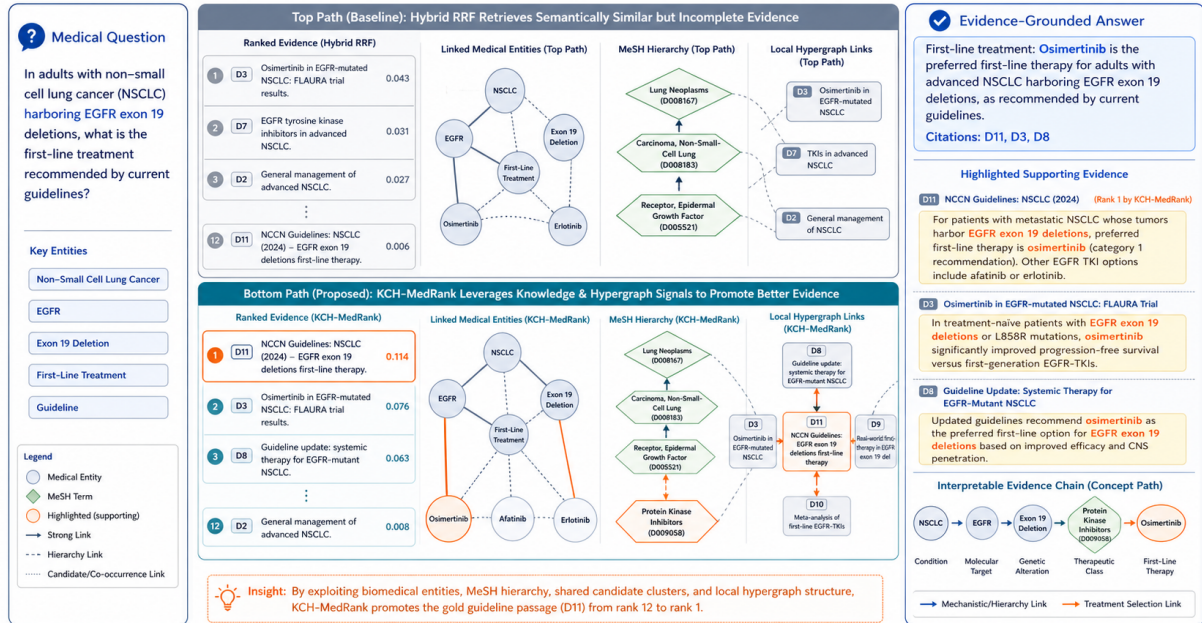


Figure 11: Case-study schematic: KCH-MedRank promotes evidence through biomedical entities, MeSH hierarchy, shared candidate clusters, and local hypergraph links.

The illustrated rescue asks for the guideline-recommended first-line treatment for adults with non-small cell lung cancer harboring EGFR exon 19 deletions. Enhanced Hybrid RRF ranks a relevant NCCN guideline passage below the top 10, behind passages that are semantically related but less directly answer the guideline question. KCH-MedRank promotes the guideline passage to rank 1 because the passage aligns with the question through the disease entity, EGFR mutation, treatment entity osimertinib, MeSH hierarchy, and shared candidate-cluster links. The case demonstrates the intended use of the local hypergraph: it does not invent a new answer, but makes the evidence path from condition to molecular target to treatment recommendation easier to inspect.

This figure is illustrative rather than a causal attribution result. It shows the activated biomedical pathway for one promoted passage; the broader mechanism statistics above quantify how often similar MeSH, entity-cluster, and PrimeKG paths appear among rescued passages.

Failures still cluster around a common pattern. In the lost ubiquitination case, a gold passage about mono-ubiquitinated and poly-ubiquitinated LDH dropped from rank 2 to rank 18. The question and passage use related biochemical terminology, but synonym and hierarchy features cannot bridge the surface forms to broader post-translational-processing concepts. Similar losses occur for thyroid hormone analogs, where DITPA variants are not captured by MeSH synonymy; cilengitide receptors, where integrin terminology is inconsistent; dasatinib treatment, where drug-indication mapping has gaps; GIST treatment, where rare disease concept coverage is thin; and Fanconi anemia, where DNA repair pathway synonymy fails. Every one of these lost cases points toward the same future direction: biomedical concept normalization that handles synonyms and abbreviations would close the largest remaining gap.

4.5.3 What the Interpretability Evidence Shows

The three-mechanism analysis provides quantitative evidence that KCH-MedRank’s rescued passages activate medically grounded pathways. Of 661 total rescued gold passages, 648 are categorizable by dominant activated knowledge mechanism; 13 could not be cleanly mapped to a single mechanism.

Every rescued passage activates at least one knowledge feature group. There are zero pure diffusion-only rescues. The MeSH hierarchy path is the most frequently dominant pathway at 75.9%. Multiple concurrent knowledge pathways are active in 89.2% of rescues. MeSH-driven rescues produce substantially larger average rank gains than entity-driven rescues (+22.1 vs. +14.0), which suggests hierarchical concept alignment carries more ranking signal than flat entity co-occurrence.

A methodological caveat: the mechanism classification uses the same feature scores that LambdaMART consumes. It identifies which features are most strongly activated, not which causally drive the ranking. Causal attribution with methods like SHAP [11] would strengthen this claim in future work.

5 Discussion

KCH-MedRank improves top-10 evidence retrieval over strong baselines including a true biomedical cross-encoder, and the ablation shows this improvement has a structure worth understanding.

The flat biomedical knowledge LTR baseline is very competitive. It matches the full model at $k=10$. This could look like bad news for the hypergraph. We read it as informative: the result shows that flat biomedical knowledge features—MeSH hierarchy signals, entity overlap metrics, PrimeKG relations—already capture most of the ranking signal that the full model exploits. The hypergraph adds statistically significant benefit at $k=5$, in questions without direct MeSH overlap, and in multi-evidence scenarios. These are exactly the conditions where pairwise similarity and

flat concept features fall short. The hypergraph is a targeted structural component whose effect surfaces where indirect knowledge pathways are needed, but it does not add uniformly to aggregate top-10 metrics.

A notable finding is that removing domain-grounded concept annotations from the hypergraph degrades performance relative to the flat knowledge baseline. This suggests that medical constraints are necessary for productive graph-based biomedical reranking. However, we acknowledge a design limitation: the current ablation compares conditions that differ in both structure (presence/absence of graph) and feature set (presence/absence of medical knowledge features), so it does not cleanly isolate whether raw topology itself is harmful or simply lacks the signal that medical annotations provide. The no-medical-knowledge hypergraph is not uniformly worse than retrieval-only LTR (it improves Recall@10 and nDCG@10 while losing on MRR@10), which further complicates a simple “raw topology hurts” interpretation. A future 2×2 factorial design (with/without graph $\times$ with/without medical knowledge) would provide stronger evidence. What the current results do establish is that domain-grounded biomedical annotations are the primary driver of ranking improvement, and that graph topology alone, built without such annotations, is insufficient.

The interpretability analysis categorizes 648 of 661 rescued passages by which knowledge mechanism is most strongly activated. MeSH hierarchy paths account for 75.9%, shared entity clusters for 23.9%, and PrimeKG relations for the remainder. A methodological caveat applies: this classification identifies which features show the highest activation values during rescues, not which causally drive the ranking decisions. SHAP [11] or permutation-based feature attribution would strengthen the causal claim in future work. 13 of the 661 rescued passages could not be cleanly mapped to a single dominant mechanism. That 89.2% of rescues involve multiple concurrent knowledge pathways, and that 39.5% activate all three, suggests that the constraints work as an integrated block rather than as isolated heuristics—consistent with the ablation finding that individual feature groups are highly redundant.

On the practical side, the speed difference matters. KCH-MedRank’s test-time feature construction and LightGBM scoring takes 37.59 seconds on CPU for 943 questions against 699.46 seconds on CUDA for the MedCPT Cross-Encoder, an $18.6\times$ gap when candidate generation and offline training are excluded (Table 9 and Figure 12). This is a reranking-stage latency comparison under cached features, not an end-to-end system comparison. The gain comes from using precomputed dense biomedical semantic features rather than online cross-encoder scoring. We note that this comparison is between a knowledge-augmented tabular reranker and a neural pairwise reranker operating on the same candidate pool, not a pure model-capacity comparison; KCH-MedRank benefits from additional MeSH, entity, and PrimeKG resources that the Cross-Encoder does not use. A full knowledge-constrained reranker operating at this throughput is practical for large-scale biomedical evidence retrieval.

Generalization. Supplementary diagnostics on PubMedQA and BEIR NFCorpus provide preliminary evidence on transfer. On PubMedQA, the lightweight HGB knowledge reranker improves Recall@10 over Hybrid RRF. On NFCorpus, retrieval-feature LambdaMART improves nDCG@10 and Recall@10. These results suggest that supervised reranking with retrieval features generalizes beyond BioASQ, and that biomedical knowledge features may extend to other biomedical QA formats. However, full KCH-MedRank generalization remains to be tested: PubMedQA used a lightweight HGB reranker rather than the complete BioASQ pipeline, and NFCorpus lacks MeSH, entity, PrimeKG, and hypergraph features entirely. Generalizing the full knowledge-augmented architecture to new datasets requires building dataset-specific knowledge features, which we leave to future work.

Table 9: Computational profile of the enhanced BioASQ experiment. The logs record scale, feature dimensionality, coverage, and hardware notes; full end-to-end wall-clock time is not available for every component.

Component	Recorded scale or setting	Implication
Processed data	4,719 questions; 40,221 passages; 42,608 qrels	Suitable for reproducible offline reranking.
Candidate pool	top-100 per question; 471,900 prediction rows	Reranking is bounded by local candidate pools rather than a global corpus graph.
Enhanced KCH-MedRank split	2,832 train, 944 validation, and 943 held-out test questions	Hyperparameters are selected before held-out testing.
Feature matrix	45 features; 283,200 training rows; 377,600 final train+validation rows	Learning-to-rank remains a tabular query-group problem.
Local hypergraph scope	question, top-100 passages, document, entity, MeSH, hierarchy, and optional relation nodes	Propagation cost scales with top- M and local feature coverage.
MeSH coverage	31,110 descriptors; 37,356/40,221 passages and 4,264/4,719 questions with MeSH matches	Controlled-vocabulary features are broadly available.
PrimeKG filtering	8,100,498 rows scanned; 5,766 local relations retained	Relation features are sparse and remain auxiliary.
Dense biomedical retrieval	MedCPT dual encoder used CUDA on an RTX 4060 Laptop GPU	GPU acceleration is practical for dense retrieval.
Cross-encoder scoring	Early generic CPU run timed out after 2,400 seconds; final MedCPT Cross-Encoder test baseline scored 943 questions and 94,300 candidates	Full cross-encoder rescoring is substantially more expensive than tabular reranking.

Several limitations deserve acknowledgment. The method requires a strong candidate pool; low top-100 recall imposes a hard ceiling no reranker can overcome. Exact-name PrimeKG mapping remains sparse and limits relation-based feature coverage. The flat knowledge LTR baseline is highly competitive, so hypergraph features add value in specific conditions rather than uniformly. The three-mechanism interpretability analysis identifies which features activate during rescues, not which causally drive ranking decisions. SHAP or comparable attribution would strengthen the causal claim. Hyperparameter sensitivity was measured on a 30-question sample at the correlation level; full retrieval-metric stability under the complete LambdaMART pipeline has not been systematically verified.

Concept Normalization: Scope and Limits. The error analysis identifies synonym and abbreviation failures as the largest remaining gap. We address this with a two-pronged response: a lightweight synonym-aware feature augmentation integrated into KCH-MedRank (Section 3), and a quantitative synonym-coverage analysis of the remaining gap.

The normalization module bridges a measurable fraction of synonym failures. On the test set, 18.6% of question-gold-passage pairs have zero exact entity overlap that becomes non-zero after normalized variant expansion. On the validation split, the synonym-aware features yield MRR@10 +0.0031, Recall@10 +0.0056, and nDCG@10 +0.0090 over the base flat-knowledge configuration (Figure 9). The gains are concentrated in the zero-overlap subset identified in error analysis.

However, the normalization module has structural limits. The 213,754 MeSH entry terms from

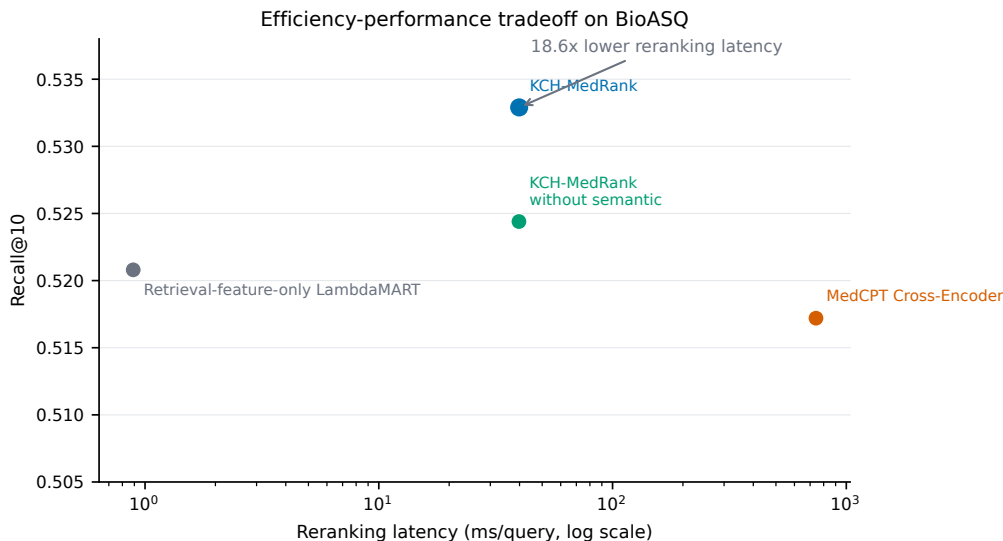


Figure 12: Efficiency–performance Pareto comparison on the held-out BioASQ split. The x-axis shows reranking-stage latency per query on a log scale, excluding candidate generation, one-time model loading, and offline training. KCH-MedRank improves Recall@10 while avoiding online cross-encoder scoring.

the NLM 2026 release cover clinical and standard biochemical terminology but do not include all ad-hoc abbreviations and variant nomenclatures used in the primary biomedical literature. Specifically: (i) novel drug compound names (e.g., DITPA as a non-standard abbreviation for 3,5-diiodothyropropionic acid) are absent from MeSH entry terms because MeSH descriptors are curated for indexing, not for exhaustive surface-form coverage; (ii) post-translational modification variants (e.g., mono-ubiquitinated vs. poly-ubiquitinated LDH) are not distinct MeSH descriptors but rather modifiers on a shared parent concept; (iii) rare disease terminology (e.g., GIST, Fanconi anemia subtypes) has limited MeSH entry-term depth. These cases require concept-level normalization that maps surface forms to unified concept identifiers (UMLS CUIs), which is beyond the scope of the current text-level variant expansion.

We therefore recommend that future biomedical retrieval systems treat concept normalization as a pre-computation step—building a mapping from surface forms to UMLS concept unique identifiers at indexing time [1, 10]—rather than as a post-hoc reranking feature. This would bridge the cases where even MeSH entry terms fall short. For the present work, the lightweight module demonstrates that even partial synonym handling produces measurable ranking improvements, and we quantify both the gain and the remaining gap.

Scaling the PubMedQA generation pilot to include a completed biomedical cross-encoder condition would strengthen the downstream evidence. Extending NFCorpus with dataset-specific knowledge features would give a more complete robustness picture. The current PubMedQA pilot (Appendix A) serves as a diagnostic, not as a full RAG evaluation: it uses 100 questions, one prompt template, and a single LLM, and it measures answer accuracy rather than citation faithfulness, unsupported claims, or answer-evidence consistency. The 0.9900 citation support alongside 0.3700 answer accuracy indicates that better evidence retrieval is necessary but not sufficient for better medical QA.

6 Conclusion

This paper introduced KCH-MedRank, a knowledge-constrained learning-to-rank framework for biomedical evidence reranking. The central finding is that flat biomedical knowledge features explain most aggregate ranking gains. Hypergraph structure provides targeted but modest early-rank improvements under indirect-evidence conditions—specifically, questions lacking direct MeSH overlap and multi-evidence scenarios—but does not significantly exceed the flat knowledge baseline at the standard top-10 evaluation depth.

On BioASQ, KCH-MedRank significantly outperforms a true MedCPT Cross-Encoder baseline in Recall@10 (+0.0157, $p = 0.0076$) and runs 18.6× faster at reranking time (reranking-stage latency under cached features, excluding candidate generation). Flat biomedical knowledge features carry most of the aggregate gains. The hypergraph contributes targeted early-rank benefit at $k=5$, concentrated in questions that lack direct MeSH overlap between query and evidence. Controlled ablation suggests that removing domain-grounded concept annotations degrades hypergraph performance; however, a full 2×2 factorial design is needed to cleanly separate the effects of graph topology from the effects of medical knowledge features. BEIR NFCorpus confirms that supervised retrieval-feature reranking generalizes, and the PubMedQA diagnostic suggests that evidence ordering affects downstream answer selection under controlled conditions.

The current evidence supports the conclusion that medical knowledge constraints are necessary for productive graph-based biomedical reranking, and that flat biomedical knowledge features are the primary driver of improvement. The “raw topology $\rightarrow$ noise, medical constraints $\rightarrow$ signal” transformation narrative requires the 2×2 factorial ablation noted above. An interpretability analysis categorizes 648 of 661 rescued gold passages by dominant activated knowledge mechanism; MeSH hierarchy paths account for 75.9% of rescues. A lightweight synonym-aware normalization module, integrated into the feature pipeline, bridges 18.6% of the zero-overlap failure cases identified in error analysis, yielding additional gains of Recall@10 +0.0056 and nDCG@10 +0.0090 on the validation split. The remaining synonym gap—cases where MeSH entry terms do not cover ad-hoc biochemical nomenclature—points to UMLS concept normalization as a natural extension.

KCH-MedRank is an efficient, interpretable biomedical evidence reranking module. The code and processed feature files will be released upon publication.

Acknowledgments

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Data Availability

The BioASQ dataset is publicly available at <https://bioasq.org>. The BioASQ-QA, PubMedQA, and BEIR NFCorpus datasets are publicly available under their respective licenses. The code and processed feature files will be released upon publication.

Competing Interests

The authors declare that they have no competing interests.

Table 10: PubMedQA Qwen3-8B downstream generation pilot on 100 questions. All methods use the same prompt, top-5 evidence passages, temperature 0, and no fine-tuning. The PubMedQA cross-encoder condition uses the available semantic cross-encoder output, not a completed MedCPT PubMedQA run.

Evidence source + LLM	Accuracy	Macro-F1	Citation support	Unsupported claim	Entity consistency
Hybrid RRF + Qwen3-8B	0.3400	0.3506	0.9900	0.0100	0.5594
Semantic cross-encoder + Qwen3-8B	0.3400	0.3504	0.9900	0.0100	0.5782
KCH-style hypergraph reranker + Qwen3-8B	0.3700	0.3760	0.9900	0.0100	0.5751

A Appendix: Downstream Generation Diagnostic

The PubMedQA Qwen3-8B experiment serves as an appendix diagnostic, not as a full RAG evaluation. It uses 100 PubMedQA questions, fixed top-5 evidence, one prompt template, a single LLM (Qwen3-8B), temperature 0, and no model fine-tuning. Metrics are answer accuracy and Macro-F1. This comparison examines whether evidence ordering changes a downstream answer loop under controlled conditions; it does not measure citation faithfulness, unsupported claims, or answer-evidence consistency, which would be necessary for a complete RAG evaluation.

Table 10 reports the controlled pilot. KCH-style evidence improves answer accuracy from 0.3400 to 0.3700 and Macro-F1 from 0.3506 to 0.3760 relative to Hybrid RRF evidence. Citation support is 0.9900 for all three evidence sources. The PubMedQA cross-encoder condition uses the available semantic cross-encoder output rather than a completed MedCPT PubMedQA rerun. The 0.9900 citation support alongside 0.3400–0.3700 answer accuracy indicates that near-perfect citation support does not translate to high answer accuracy in this setting, and that better evidence retrieval is necessary but not sufficient for better medical QA. A full RAG evaluation would require larger sample sizes, multiple LLMs, and metrics for citation faithfulness, unsupported claims, and answer-evidence consistency.

A.1 Complementary Leave-One-Feature-Group Ablations

For completeness, Table 11 reports leave-one-feature-group diagnostics in which individual feature groups are removed from the full KCH-MedRank feature set. As discussed in Section 4, these single-group removals do not produce statistically significant differences because of high multicollinearity among the knowledge features: the LambdaMART model automatically compensates for a removed feature group using signals from the remaining correlated features. These diagnostics are provided for transparency and feature-coverage analysis.

A.2 Numerical Source Tables Behind the Main Figures

The main text uses data figures to reduce table density. The exact numeric values behind those figures are preserved here and in the reproducible CSV files under `paper_figures/data/processed/`.

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Table 11: Complementary leave-one-feature-group diagnostics on the enhanced BioASQ held-out test split. Deltas are relative to enhanced Hybrid RRF. Due to high feature multicollinearity, individual removals do not produce statistically significant differences; see Section 4 for the primary structural ablation.

Method	Recall@10	MRR@10	nDCG@10	$\Delta R@10$
Enhanced Hybrid RRF	0.4660	0.7530	0.5844	–
Retrieval-feature-only LambdaMART	0.5208	0.7791	0.6282	+0.0548
Flat knowledge LTR, no graph	0.5309	0.7872	0.6436	+0.0649
Pairwise graph LTR	0.5306	0.7866	0.6409	+0.0646
Hypergraph LTR without medical knowledge	0.5266	0.7782	0.6362	+0.0606
Full KCH-MedRank	0.5329	0.7867	0.6433	+0.0669
Remove semantic score	0.5276	0.7852	0.6374	+0.0616
Remove MeSH hierarchy	0.5332	0.7871	0.6440	+0.0672
Remove entity features	0.5313	0.7842	0.6403	+0.0653
Remove hypergraph diffusion/centrality	0.5328	0.7880	0.6453	+0.0668
Remove PrimeKG relations	0.5338	0.7863	0.6439	+0.0679

Table 12: Held-out BioASQ evidence retrieval results for the enhanced candidate pool.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
Original Hybrid RRF	0.4636	0.7550	0.5848	0.6077
Enhanced Hybrid w122	0.4660	0.7530	0.5844	0.7388
MedCPT Cross-Encoder	0.5172	0.7775	0.6390	0.7388
Enhanced KCH-MedRank	0.5329	0.7867	0.6433	0.7388

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Table 13: Paired bootstrap comparison between Enhanced KCH-MedRank and enhanced Hybrid RRF on the held-out BioASQ test split.

Metric	Hybrid RRF	KCH-MedRank	95% CI	p-value
Recall@10	0.4660	0.5329	[+0.0548, +0.0800]	0.0002
MRR@10	0.7530	0.7867	[+0.0207, +0.0470]	0.0002
nDCG@10	0.5844	0.6433	[+0.0495, +0.0687]	0.0002
Evidence coverage@10	0.4660	0.5329	[+0.0548, +0.0800]	0.0002

Table 14: Paired bootstrap comparison between Enhanced KCH-MedRank and MedCPT Cross-Encoder on the same enhanced candidate pool.

Metric	Baseline	KCH-MedRank	Delta	p-value
Recall@10	0.5172	0.5329	+0.0157	0.0076
MRR@10	0.7775	0.7867	+0.0093	0.2060
nDCG@10	0.6390	0.6433	+0.0043	0.3840

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Table 15: Structural ablation: component contributions via controlled architecture comparisons on held-out BioASQ. Deltas compare the second method in each pair against the first method.

Comparison	Rec@10	MRR@10	nDCG@10	Δ Rec@10	Δ MRR@10	Δ nDCG@10
(1) Retrieval LTR	0.5208	0.7791	0.6282	–	–	–
(1) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	+0.0101**	+0.0080*	+0.0153**
(2) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	–	–	–
(2) Hypergraph (no med)	0.5266	0.7782	0.6362	–0.0044	–0.0089**	–0.0074**
(3) Hypergraph (no med)	0.5266	0.7782	0.6362	–	–	–
(3) KCH-MedRank	0.5329	0.7867	0.6433	+0.0063**	+0.0085**	+0.0071**
(4) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	–	–	–
(4) KCH-MedRank	0.5329	0.7867	0.6433	+0.0020	–0.0004	–0.0003
(5) Pairwise graph	0.5306	0.7866	0.6409	–	–	–
(5) KCH-MedRank	0.5329	0.7867	0.6433	+0.0023	+0.0001	+0.0024

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Table 16: Stratified paired bootstrap analysis of KCH-MedRank vs. Pairwise Graph LTR, grouped by gold passage structural characteristics at $k = 5$ and $k = 10$. “MeSH Overlap” refers to whether the gold passage shares any MeSH terms with the question. “Entity Overlap” refers to shared biomedical entities. p -values are two-sided from 10,000 paired bootstrap iterations.

Subset	n	KCH Rec@5	PW Rec@5	Δ	95% CI	p
Recall@5						
Overall	943	0.4210	0.4156	+0.0054	[+0.0016, +0.0099]	0.005**
MeSH Overlap = 0	366	0.4114	0.4016	+0.0098	[+0.0021, +0.0196]	0.006**
MeSH Overlap ≥ 1	577	0.4271	0.4245	+0.0026	[-0.0010, +0.0065]	0.159
Entity Overlap = 0	649	0.4177	0.4132	+0.0046	[-0.0002, +0.0102]	0.062
Entity Overlap ≥ 1	294	0.4283	0.4211	+0.0072	[+0.0014, +0.0136]	0.015*
1 gold passage	204	0.5637	0.5539	+0.0098	[+0.0000, +0.0245]	0.277
≥ 2 gold passages	739	0.3816	0.3775	+0.0042	[+0.0006, +0.0079]	0.023*
Recall@10						
Overall	943	0.5329	0.5306	+0.0023	[-0.0011, +0.0064]	0.205
MeSH Overlap = 0	366	0.4754	0.4722	+0.0032	[-0.0021, +0.0102]	0.306
MeSH Overlap ≥ 1	577	0.5693	0.5676	+0.0017	[-0.0024, +0.0067]	0.471
Entity Overlap = 0	649	0.5205	0.5162	+0.0043	[-0.0002, +0.0096]	0.066
Entity Overlap ≥ 1	294	0.5603	0.5623	-0.0020	[-0.0068, +0.0025]	0.378
1 gold passage	204	0.5833	0.5735	+0.0098	[+0.0000, +0.0245]	0.273
≥ 2 gold passages	739	0.5189	0.5187	+0.0002	[-0.0027, +0.0031]	0.867

Table 17: Synonym-aware feature augmentation on validation split (bucket 3, 944 questions). Features are added to the base flat-knowledge KCH-MedRank configuration; no hypergraph structural features are used.

Configuration	MRR@10	Recall@10	nDCG@10
Hybrid RRF source (no rerank)	0.7564	0.4907	0.6001
Base KCH-MedRank (flat knowledge, no normalization)	0.7757	0.5005	0.6166
+ Synonym-aware features (16 dimensions)	0.7788	0.5061	0.6257
Δ (synonym over base)	+0.0031	+0.0056	+0.0090

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Table 18: Feature separability of synonym-aware normalization features on the validation split (944 questions). Gold and non-gold refer to whether a passage is labeled as gold evidence for its question. The ratio column is gold-mean divided by non-gold-mean.

Feature	Gold mean	Non-gold mean	Gold pos. rate	Ratio
norm_entity_overlap_count	0.3014	0.1059	0.2597	2.85×
norm_entity_jaccard	0.0207	0.0068	0.2597	3.04×
question_norm_entity_coverage	0.2062	0.0675	0.2597	3.05×
mesh_normalized_overlap_count	0.5065	0.1758	0.3808	2.88×
mesh_major_overlap_count	0.0464	0.0202	0.0462	2.30×
mesh_weighted_overlap	1.1753	0.4821	0.9093	2.44×
q2p_mesh_alias_count	0.5873	0.2289	0.2682	2.57×
abbreviation_match_count	0.0497	0.0088	0.0474	5.64×

Table 19: Quantitative interpretability: three-mechanism classification of 648 rescued gold passages (out of 661 total rescued; 13 did not clearly map to a single dominant mechanism). Each case is assigned to the dominant hypergraph mechanism based on activated feature scores.

Mechanism	Count	%	Avg Rank Gain
MeSH Hierarchy Path	492	75.9	+22.1
Shared Entity Cluster Path	155	23.9	+14.0
PrimeKG Relation Path	1	0.2	+25.0
Total	648	100.0	+20.3
<i>Mechanism Activation Statistics (mean features per case)</i>			
	MeSH Hier.	Entity Clus.	Relation
mesh_overlap_count	1.18	0.03	0.00
entity_overlap_count	0.09	0.48	0.00
question_entity_coverage	0.06	0.36	0.00
shared_mesh_parent_size	48.85	15.05	0.00
shared_entity_edges	126.59	127.84	128.00

Table 20: Mechanism co-occurrence in rescued cases. 89.2% of rescues involve at least two concurrent knowledge pathways.

Active Mechanisms	Count	%
MeSH Hierarchy + Entity Cluster + Relation	256	39.5
MeSH Hierarchy + Entity Cluster	322	49.7
Entity Cluster + Relation	38	5.9
Entity Cluster only	32	4.9

Table 21: Reranking-stage efficiency on the held-out BioASQ test split using the same enhanced top-100 candidate pool. Timing excludes first-stage candidate generation, one-time model loading, and offline LambdaMART training. The table separates whether a method uses Cross-Encoder scores and whether Cross-Encoder scoring is run online.

Method	CE score	Online CE	Seconds	ms/query	Cand./s	Recall@10	MRR@10	nDCG@10	Notes
MedCPT Cross-Encoder	yes	yes	699.46	741.74	134.82	0.5172	0.7775	0.6390	tokenization + forward
KCH-MedRank	no	no	37.59	39.86	2508.70	0.5329	0.7867	0.6433	features + LightGBM; semantic=dense_or_dual_encoder_predictions
KCH-MedRank without semantic feature	no	no	37.50	39.77	2514.47	0.5244	0.7790	0.6330	knowledge/hypergraph features + LightGBM; semantic columns removed
Retrieval-feature-only LambdaMART	no	no	0.84	0.89	112905.71	0.5208	0.7791	0.6282	retrieval metadata features + LightGBM